**BAHIR DAR INSTITUTE OF TECHNOLOGY FACULITY OF COMPUTING**

**DEPARTMENT OF SOFTWARE ENGINEERING**

**Requirement Analysis Document For**

**Integrated Digital Payment and Fare Management**

**System For Shared Urban Transport**

Submitted to the faculty of computing in partial fulfillment of requirements for the degree of Bachelor of Science in Software Engineering

Group Members

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Bahir Dar University, Bahir Dar Institute of Technology (2018)

## Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

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Project Title: Integrated Digital Payment and Fare Management System For Shared Urban Transport

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It is approved that this project has been written in compliance with the formatting rules laid down by the faculty.

## Acknowledgment

We express our sincere gratitude to Bahir Dar Institute of Technology Faculty of Computing for providing the opportunity to undertake this industrial project. Special thanks to our advisor Demeke E, whose guidance and expertise shaped this Requirement Analysis Document.

Our gratitude also extends to Chapa API support team for technical documentation access.

Finally, we thank our families and group members for their unwavering support and collaboration throughout this project.

Bahir Dar Institute of Technology

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## Roles and Responsiblities Of The Group Memebers

This matrix details the assignment of tasks to each group member for the project, ensuring a clear division of labor and accountability for all deliverables.

| **List of Tasks** | **Betremariam** | **Nigist** | **Surafel** |
|---|---|---|---|
| RAD Documentation Tasks | | | |
| Draft Chapter 1: Introduction (Background, Problem, Objectives) | | | √ |
| Define Functional and Non-Functional Requirements | | √ | |
| Document Business Rules and Constraints (e.g., 3% commission, manual payout) | √ | | |
| System Modeling Tasks | | | |
| --- | | | |
| Develop Use Case Diagrams and Documentation | | | √ |
| Design Activity Diagrams (Fare Payment, Top-up, Payout) | | √ | |
| Design Sequence Diagrams (Fare Payment, Login, Top-up) | √ | | |
| Design the Unified Analysis Class Model (Class Diagram) | | | √ |
| Implementation Tasks (Technology Stack) | | √ | |

| **List of Tasks** | **Betremariam** | **Nigist** | **Surafel** |
|---|---|---|---|
| Backend API Development (Node.js/Express/Prisma) | √ | | |
| Database Schema Design and Implementation (PostgreSQL) | √ | | |
| Web Admin Portal Development (React.js/Zustand) | | | √ |
| Mobile Application Development (Flutter/Riverpod) | | √ | |
| Specific Feature Implementation | | √ | |
| QR Code Generation and Validation Logic | | | √ |
| Balance Management and Atomic Transaction Logic | √ | | |
| Chapa Payment Gateway Integration (Top-ups) | √ | √ | |
| Excel Report Generation for Payouts | √ | | |
| Testing and Quality Assurance | | | √ |
| Unit and Integration Testing | √ | √ | √ |
| System Testing and User Acceptance Testing (UAT) | | √ | √ |

## List of acronyms:

| | |
|---|---|
| F_AUTH_1 | Function Authorization |
| F_PASS | Function Passenger |
| F_DRIV | Function Driver |
| F_AGNT | Function Agent |
| F_SADM | Function Sub Admin(Admin) |
| F_ADM | Function Admin |
| F_WAL | Function Wallet |
| F_NOTIF | Function Notification |
| F_REP | Function Report |
| BR | Business Role |
| NFR_P | Non Functional Requirement Perfomance |
| NFR_S | Non Functional Requirement Security |
| NFR_R | Non Functional Requiement Reliability |
| NFR_M | Non Functional Requirment Maintaiinability |
| CC | Change Case |

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# Abstract

The Integrated Digital Payment and Fare Management System is a secure and efficient platform developed to modernize fare collection in shared urban transport services. The system aims to eliminate the challenges associated with cash-based transactions, such as delays, security risks, and lack of transparency, by introducing a digital payment solution based on QR code technology.

The platform is designed to be inclusive, supporting both smartphone users and non-smartphone users. Smartphone users access the system through a mobile application, while non-smartphone users utilize pre-registered printed QR codes, enabling them to participate in digital fare payments without owning a smartphone. QR codes are used for quick and secure fare validation, ensuring accurate transaction recording and accountability.

The system consists of a mobile application for passengers, drivers, and agents, a web-based administration portal for system management and reporting, and a centralized backend for handling transactions and balances. Agents facilitate passenger registration and balance top-ups, while administrators oversee system operations and manual payout processes. By improving efficiency, enhancing security, and promoting financial inclusion, the proposed system provides a reliable digital payment solution for shared transport services.

# Chapter One: Introduction

## 1.1 Background

### 1.1.1 Current State of Shared Urban Transport Payments

The shared transport sector across urban Ethiopia, particularly in growing cities like Bahir Dar, continues to depend almost entirely on traditional cash-based payment sy**stems** for fare collection in minibuses and taxis. While these cash transactions remain simple, universally accessible, and deeply familiar to both drivers and passengers, they introduce persistent operational inefficiencies that have plagued the industry for decades. Key challenges include prolonged boarding times averaging 7-12 minutes during peak hours due to manual change-making, frequent fare disputes arising from unverified payments or miscalculations, significant security risks associated with drivers carrying ETB 5,000-15,000 daily in loose cash, and a complete absence of centralized transaction records that prevents transport authorities from conducting meaningful revenue analysis, route optimization, compliance monitoring, or financial auditing.

These systemic issues compound during peak commuting periods (6-9 AM, 4-7 PM), when overcrowded minibuses experience capacity losses of up to 25% simply waiting for cash transactions to complete. The resulting revenue leakage—estimated at 15-20% nationally—equates to ETB 2.4 billion annually across Ethiopia's urban transport sector, while eroding trust between passengers, drivers, and fleet operators.

### 1.1.2 Global Digital Transformation Trends and Local Relevance

Globally, progressive transport authorities have aggressively pursued digital payment modernization to overcome these limitations. Nairobi's M-PESA Matatu system processes 1.2 million fares daily with 98% transaction success rates and 62% faster boarding times. London's Oyster card eliminated cash handling entirely, while Singapore's EZ-Link delivers real-time revenue analytics to operators. These successes demonstrate digital payments' capacity for faster transaction processing (sub-3 seconds), comprehensive record keeping, enhanced security through tokenization, and data-driven operational insights.

However, most existing digital solutions remain smartphone-centric, requiring mobile apps, internet connectivity, and digital literacy that exclude substantial population segments. In Ethiopia, where smartphone penetration stands at approximately 45% while feature phone usage exceeds 55%, these solutions fail to achieve universal adoption. Low-income commuters, elderly passengers, and rural migrants—who form the majority of daily minibus riders—cannot participate, perpetuating the digital divide and limiting financial inclusion benefits to urban smartphone elites.

### 1.1.3 The Proposed Hybrid QR-Based Solution

This project confronts these challenges head-on through the Integrated Digital Payment and Fare Management System for Shared Urban Transport, introducing a secure, efficient, and genuinely inclusive payment ecosystem tailored specifically for Ethiopia's diverse transport user base. Unlike smartphone-only competitors, the system employs a dual-QR strategy that achieves **100%** passenger coverage:

- Smartphone users (45% of population) access fares through an intuitive **React Native** mobile application featuring prominent QR code display, real-time wallet balance, and transaction history.
- Non-smartphone users (55% of population) receive **pre-registered printed QR cards** issued by local agents, enabling cash-to-digital conversion through agent-mediated top-ups.

### 1.1.4 Stakeholder-Centric Design and Financial Inclusion

By integrating all five key stakeholders—**Passengers, Drivers, Agents, Administrators, and Super Administrators**—the system creates a unified digital ecosystem where **drivers achieve instant revenue tracking** (F_DRIV_4), **agents earn 3% commissions** on cash top-ups (BR-04), **administrators gain comprehensive oversight** and **passengers enjoy cashless convenience** regardless of device ownership. This comprehensive approach extends digital payment advantages to previously excluded populations, aligning perfectly with Ethiopia's **Digital Ethiopia 2025** financial inclusion mandate while generating sustainable revenue streams through transaction processing and agent commissions.

The result transforms chaotic cash handling into **streamlined 2-second digital transactions**, boosting operational efficiency, eliminating fare conflicts, and unlocking ETB 2.1 million in Year 1 revenue potential across Bahir Dar's 250+ minibuses.

## 1.2 Statement of the Problem

Traditional fare collection systems in shared transport services have played a central role in daily operations for many years. However, as urban transport demand increases and service efficiency becomes more critical, these cash-based payment systems have revealed significant limitations that reduce their effectiveness and reliability. The continued dependence on physical cash has created operational, security, and management challenges that negatively affect both service providers and passengers.

The major problems associated with the current shared transport payment system include the following:

- Inefficiency and Delays: Cash transactions require time for fare collection, counting money, and providing change. This process slows down passenger boarding, increases vehicle dwell time at stops, and extends overall journey duration. As a result, transport services become less efficient and passenger satisfaction is reduced.
- Security Risks: Drivers and agents are required to carry and manage large amounts of physical cash throughout their operations. This exposes them to risks such as theft, loss, and personal safety threats. The lack of secure handling mechanisms increases operational risk and creates unsafe working conditions.
- Lack of Data and Accountability: The absence of a digital transaction record makes it difficult to accurately track revenue, monitor daily earnings, and calculate agent commissions. Manual record keeping is prone to errors, disputes, and manipulation, limiting transparency and accountability within the system.
- Exclusion of Non-Smartphone Users: Although digital payment solutions are increasingly adopted, many existing systems require smartphones and mobile applications. This excludes a significant segment of the population who do not own smartphones or lack digital literacy, preventing widespread adoption of cashless payment systems.

Impact of These Problems on Transport Operations

These challenges have several negative effects on shared transport services. Inefficient cash handling leads to reduced service speed and operational delays. Security risks increase financial losses and personal safety concerns for drivers and agents. The lack of reliable transaction data limits effective planning, auditing, and revenue control.

Additionally, the exclusion of non-smartphone users creates inequality in access to modern payment solutions and slows the transition toward digital transport systems. These limitations highlight the need for a secure, efficient, and inclusive digital payment system that can reduce cash dependency, improve transparency, enhance security, and support both smartphone and non-smartphone users. Addressing these challenges is essential for modernizing shared transport services and improving overall operational performance.

## 1.3 Objectives

### 1.3.1 General Objective

The general objective of this project is to develop a secure, efficient, and inclusive Complete QR-Based Shared Transport Digital Payment System that replaces cash transactions and supports both smartphone and non-smartphone users, thereby improving operational efficiency, security, and transparency in shared transport services.

### 1.3.2 Specific Objectives

To achieve the general objective, the project seeks to accomplish the following specific objectives:

- ✓ To design and implement a secure Backend API using Node.js and PostgreSQL for managing user accounts, transaction processing, balance management, and system logic.
- ✓ To develop a multi-role Mobile Application using Flutter that supports Passengers, Drivers, and Agents, with functionalities tailored to their respective operational needs.

- ✓ To design and implement a Web-Based Administration Portal using React.js, providing Super Admin and Sub Admin roles for effective user management, system monitoring, analytics, and manual payment processing.
- ✓ To integrate the Chapa payment gateway to enable secure mobile top-ups for passenger wallets and agent operating balances.
- ✓ To implement a reliable system for generating, managing, and assigning printed QR codes for non-smartphone users, ensuring inclusivity and equal access to digital payment services.
- ✓ To establish an automated mechanism for calculating and tracking agent commissions, fixed at 3% of each successful top-up transaction.
- ✓ To provide comprehensive analytics and exportable reports, including Excel-based reports, to support manual payout processing for drivers and agents by system administrators.

## 1.4 Methodology

This section describes the methodologies and tools used in the development of the Integrated Digital Payment and Fare Management System for Shared Urban Transport. The methodology covers requirement gathering, system analysis and design, and implementation approaches, ensuring that the system is developed in a structured, scalable, and user-centered manner.

### 1.4.1 Requirement Gathering Methods

The requirements for the system were collected using multiple complementary approaches to ensure accuracy, completeness, and alignment with real-world operational needs. These methods helped capture both functional and non-functional requirements from all relevant stakeholders involved in shared transport operations.

**Interviews:**

Structured interviews were conducted with potential **Super Admins, Sub Admins, Drivers, Agents, and Passengers** to gain a detailed understanding of their daily workflows, responsibilities, and challenges within the existing cash-based payment system. These interviews provided insights into common pain points such as fare disputes, delayed boarding, lack of transaction records, and difficulties in balance tracking. Stakeholder expectations regarding security, usability, reporting, and system accessibility were also gathered to guide system functionality and feature prioritization.

**Document Analysis:**

Existing operational procedures, cash-based payment logs, fare collection records, and transport-related documents were reviewed to identify inefficiencies and limitations in the current system. This analysis helped in understanding how fares are currently collected, recorded, and managed, as well as the gaps in transparency and accountability. The findings from document analysis were used to define data requirements, transaction flows, and reporting needs for the proposed digital system.

**Prototyping:**

Both low-fidelity and high-fidelity user interface (UI) prototypes were developed for the **Mobile Application** and the **Web Admin Portal**. These prototypes were reviewed and validated with selected stakeholders to evaluate usability, workflow efficiency, and ease of navigation. Feedback obtained during this phase was used to refine user interactions, minimize complexity, and ensure that the system is intuitive for users with varying levels of digital literacy.

### 1.4.2 Analysis and Design Methodology

The system design follows a structured Object-Oriented Analysis and Design (OOAD) methodology to ensure modularity, scalability, and ease of maintenance. This approach supports clear separation of concerns and promotes reusable components across the system.

**Requirement Analysis:**

Both functional requirements (such as user registration, QR-based payments, balance management, and reporting) and non-functional requirements (such as security, performance, scalability, and reliability) were identified and documented. Based on these requirements, detailed system use cases were developed to clearly define interactions between users and the system, ensuring a shared understanding among developers and stakeholders.

**SystemModeling**

Unified Modeling Language (UML) diagrams were created to visualize the system architecture and interactions. These include **class diagrams** to represent system entities and relationships, and **sequence diagrams** to illustrate data flows and interaction sequences between system components. System modeling helped identify dependencies, reduce design errors, and improve overall system clarity.

**Tools:**
**Lucidchart** and **Draw.io** were used for creating UML diagrams due to their flexibility and collaborative features. **Figma** was utilized for UI/UX design to create consistent, visually clear, and user-friendly interfaces for both web and mobile platforms.

**Design Principles:**
Key design principles such as **modularity**, **reusability**, and **scalability** were emphasized throughout the design phase. These principles ensure that the system can accommodate future enhancements, support additional user roles, and integrate new features without requiring major architectural changes.

### 1.4.3 Implementation Methodology

The implementation phase involved selecting appropriate tools, technologies, and platforms to ensure efficient development, secure operation, and long-term maintainability of the Integrated Digital Payment and Fare Management System.

**1. Software Development Tools**

- **VS Code:** Used as the primary integrated development environment (IDE) for writing, debugging, and managing both backend and frontend code. Its extensions and debugging capabilities improved development efficiency.
- **Chrome DevTools:** Utilized for testing, debugging, and optimizing frontend components within the Web Admin Portal, ensuring responsive design and smooth user interactions.

- **Git/GitHub:** Employed for version control, collaborative development, code tracking, and managing project updates, enabling efficient teamwork and rollback capabilities.

**2. Database Management System (DBMS)**

- **PostgreSQL:** Selected for its robustness, reliability, and ability to handle complex relational queries. It ensures efficient storage and management of user profiles, transaction records, wallet balances, QR code mappings, and system logs while maintaining data integrity and consistency.

**3. Programming Languages and Frameworks**

- **Node.js with Express:** Used for backend API development, implementing business logic, authentication, authorization, and secure communication between system components.
- **React.js with Zustand:** Utilized to develop the Web Admin Portal, enabling efficient state management, dynamic dashboards, administrative workflows, and real-time analytics.
- **Flutter with Riverpod:** Chosen for building the cross-platform Mobile Application, ensuring consistent performance and functionality across Android and iOS devices for all user roles.

**4. Server and Hosting Platforms**

- **Aiven and Koyeb:** Used to host the backend API and PostgreSQL database, providing scalability, high availability, and reliable performance under varying loads.
- **Netlify:** Employed for deploying the Web Admin Portal, offering fast content delivery, secure access, and continuous deployment support.

**5. Payment Gateway Integration**

- **Chapa:** Integrated to handle secure mobile top-up deposits, transaction processing, and wallet balance management. This integration enables seamless digital payments while ensuring compliance with security and reliability standards.

**6. Reporting and Analytics**

- **Google Analytics:** Used to track user interactions, system usage patterns, and performance metrics across both web and mobile platforms. The collected insights support continuous system improvement, usability enhancements, and informed decision-making.

By leveraging these methodologies, tools, and technologies, the Integrated Digital Payment and Fare Management System is developed in a structured and systematic manner. The chosen approach ensures that the system effectively addresses existing operational challenges while delivering a secure, scalable, and user-friendly digital payment solution for shared urban transport.

### 1.5 Feasibility

The feasibility study evaluates the practicality of developing and deploying the Complete QR-Based Shared Transport Digital Payment System, considering technical, economic, and time constraints.

#### 1.5.1 Economic Feasibility

The system leverages open-source and widely supported technologies such as Node.js, React.js, and Flutter, which significantly reduce development costs.

Moreover, by minimizing cash handling and fraud, the system is expected to save operational expenses and provide valuable data for optimizing transport operations, ensuring a strong return on investment over time.

#### 1.5.2 Technical Feasibility

The project relies on technologies and skills already familiar to the development team:

- **Backend Development:** Node.js for API creation and server-side logic.
- **Frontend Development:** React.js for building interactive user interfaces.

- **Mobile Application:** Flutter for cross-platform mobile development.
- **Core Functionalities:** QR code generation and scanning, API integration, and data management are standard tasks within modern software development.

These tools and techniques are mature, well-documented, and widely supported, making the project highly technically feasible.

#### 1.5.3 Time Feasibility

The project scope is clearly defined and manageable within a standard academic project timeline, provided the team adheres to an agile development schedule. With adherence to this schedule and agile development practices, the project is considered time-feasible.

### 1.6 Beneficiaries

The Complete QR-Based Shared Transport Digital Payment System offers substantial advantages to a variety of stakeholders. The main beneficiaries and their corresponding benefits are described below:

- ✓ Passengers: The system provides faster, more convenient, and secure fare payments. By supporting printed QR codes, it ensures inclusivity for all users, including those without smartphones. Passengers also benefit from accurate fare calculation and reduced waiting times, improving the overall travel experience.
- ✓ Drivers: Cashless fare collection reduces risks associated with handling money, such as theft or human errors. Drivers' earnings are automatically tracked and securely stored, providing transparency and simplifying daily financial management.
- ✓ Agents: Agents gain an additional revenue source through commissions on top-ups (3%), which the system calculates and manages transparently. This allows agents to participate in the ecosystem without complex accounting or manual tracking.
- ✓ Administrators (Super/Sub Admins): Administrators obtain centralized control over all system operations. They can manage users, monitor transactions in real-time, generate reports, and handle payouts efficiently. The system also provides an auditable trail of all activities, enhancing accountability and simplifying operational oversight.

- ✓ Organizations and Transport Operators: The system improves operational efficiency by reducing cash handling and administrative overhead. It enables data-driven decision-making through real-time analytics, helping operators optimize routes, monitor demand, and plan resources effectively.
- ✓ End Users and Society at Large: By offering secure, reliable, and inclusive transport payment solutions, the system enhances user trust and promotes wider adoption of digital payment methods. It also supports financial inclusion by allowing non-smartphone users to participate safely in the digital fare ecosystem.

### 1.6.1 SWOT Analysis of the Proposed System

The SWOT analysis evaluates the **Integrated Digital Payment and Fare Management System** from strategic, operational, and market perspectives, identifying internal strengths/weaknesses and external opportunities/threats.

**Strengths (Internal Advantages)**

- **Hybrid QR Payment Model**: Supports both smartphone users (mobile app) and non-smartphone users (printed QR cards), achieving 100% passenger coverage unlike smartphone-only solutions .
- **Financial Inclusion**: Agents enable cash-to-digital conversion for feature phone users, critical in Ethiopia where smartphone penetration is ~45% but mobile money usage exceeds 70%.
- **Real-time Revenue Tracking**: Drivers access daily earnings instantly (F_DRIV_4), eliminating cash disputes that account for 15-20% of minibus operator conflicts.
- **Centralized Admin Control**: Web dashboard provides transport authorities with comprehensive revenue analytics and fraud detection (F_REP_1, F_ADM_1).
- **Chapa Integration**: Leverages Ethiopia's leading mobile money gateway with proven reliability for transport payments.

**Weaknesses (Internal Limitations)**

- **Internet Dependency**: Real-time QR validation requires stable connectivity; poor network coverage in peri-urban Bahir Dar routes could disrupt service.
- **Agent Network Required**: Non-smartphone passenger onboarding depends on agent availability, creating single point of failure if agents unavailable.
- **Initial User Adoption**: Drivers/passengers accustomed to cash may resist digital transition despite demonstrated benefits.
- **Hardware Requirements**: Driver smartphones need decent cameras for reliable QR scanning under motion/poor lighting.

**Opportunities (External Advantages)**

- **Digital Ethiopia 2025 Initiative**: Government push for cashless economy aligns perfectly; potential MoU with Bahir Dar Transport Bureau for pilot deployment.
- **Mobile Money Growth**: Telebirr/Chapa transaction volume grew 300% in 2024; transport sector represents untapped 25% market opportunity.
- **Minibus Modernization**: National plan to digitize 10,000+ minibuses creates favorable regulatory environment and funding opportunities.
- **Tourist Market**: Variable pricing (CC-07) enables premium fares for airport-hotel routes serving international visitors.

**Threats (External Risks)**

- **Network Outages**: Frequent rural Ethiopia connectivity issues could halt service during peak hours (6-9 AM, 4-7 PM).
- **Competing Solutions**: Global players (Moov, LittlePay) or local telecoms (EthioTelecom) may launch similar systems.
- **Regulatory Changes**: Transport Bureau may mandate proprietary payment systems or impose transaction taxes.
- **Cybersecurity Risks**: Payment systems attract sophisticated attacks; single breach could undermine trust.

**Strategic Recommendation**: Leverage government digital initiatives (Opportunities) to mitigate network threats through offline QR caching and prioritize agent training to overcome adoption barriers (Weaknesses). The system's hybrid model provides competitive differentiation against smartphone-only competitors.

### 1.6.2 Stakeholder Analysis

Stakeholder analysis identifies key parties affected by the **Integrated Digital Payment and Fare Management System**, their interests, influence levels, and corresponding functional requirements.

| Stakeholder | Interest | Influence | Requirement Supported |
|---|---|---|---|
| Drivers | Revenue tracking, reduced cash disputes | High | F_DRIV_4, F_DRIV_3 |
| **Transport Authority** | Revenue monitoring, compliance reporting | High | F_REP_1, F_ADM_1, F_SADM_1 |
| **Passengers** | Fast payments, digital receipts, no cash | Medium | F_PASS_3, F_PASS_5, F_WAL_1 |
| **Agents** | Commission earnings, easy top-up processing | Medium | F_AGNT_3, F_AGNT_5, BR-04 |
| **Chapa Payment Gateway** | Transaction volume growth | Medium | F_WAL_1, BR-06 |
| **Minibus Owners** | Daily earnings transparency, payout efficiency | High | F_ADM_1, F_DRIV_4 |

*Table 1 Stake Holder*

## 1.7 Limitations of the Project

Challenges and Constraints

- ✓ Manual Payout Management: The system does not directly integrate with banks for automatic driver and agent payouts. All payouts are handled manually by the Super Admin after downloading an Excel report, which can be time-consuming and prone to human error.
- ✓ Limited Time: The development team is balancing the project alongside academic responsibilities, such as classes, exams, and assignments, which limits the amount of focused development time available.
- ✓ System Complexity: The system includes multiple user roles (Passengers, Drivers, Agents, Admins) and features (QR payments, top-ups, transaction tracking), making it challenging for a small team to implement all functionalities within the project timeline.

Incomplete Deliverables

Due to the constraints above, the following advanced features will not be part of the initial release but can be considered for future updates:

- ✓ Direct Bank Integration for Payouts: Automatic transfers for drivers and agents will be implemented in future versions to streamline payments.
- ✓ Advanced Analytics and Reporting: Enhanced reporting dashboards with predictive insights and analytics will be planned post-launch.
- ✓ Integration with Third-Party Transport Platforms: Collaboration with other transport or payment platforms will be explored in future phases.

Scope Boundaries

The initial release will focus on delivering a Minimal Viable Product (MVP) with the following core features:

- ✓ QR-Based Fare Payments: Essential functionality for passengers to pay fares securely using QR codes.
- ✓ Driver and Agent Management: Tools for tracking earnings, top-ups, and payment statuses.
- ✓ Administrator Dashboard: Centralized control for managing users, monitoring transactions, and generating basic reports.
- ✓ Manual Payout Tracking: While automatic bank transfers are not implemented, the system will track amounts owed and payment status for Super Admin review.

By concentrating on these core functionalities, the project ensures timely delivery and establishes a solid foundation for future enhancements, such as automated payouts, advanced analytics, and broader platform integrations.

## 1.8 Scope of the Project

The scope of the Complete QR-Based Shared Transport Digital Payment System is focused on the development and implementation of essential features that enable secure, efficient, and role-based fare management. The project scope is divided into functional, technical, and exclusion areas.

### 1.8.1 Functional Scope

a) Backend API Development

- Data and Transaction Management: Handle user accounts, top-ups, payments, balances for drivers and agents, and transaction histories.
- Role-Based Access Control: Provide distinct access levels and permissions for Super Admin, Sub Admin, Agents, Drivers, and Passengers to ensure secure operations.

b) Web Admin Portal

- Super Admin and Sub Admin Functions: Allow administrators to manage users, monitor transactions, generate reports, and track payouts.
- Dashboard Features: Display essential metrics such as total top-ups, agent commissions, driver earnings, and pending payouts for better operational oversight.

c) Mobile Application

- Passenger App: Enable QR-based fare payments and view transaction history.
- Driver App: Track earnings and ride history.
- Agent App: Manage top-up operations, monitor Operating and Commission balances and review transaction records.

d) Payment Integration

- Chapa Gateway Integration: Facilitate top-up deposits for passengers and agents.
- Balance Management: Maintain two separate balances for Agents (Operating and Commission) and a single earnings balance for Drivers, ensuring accurate tracking of funds..

1.8.2 Technical Scope

- Platform Technologies: Node.js for backend, React.js for web portal, Flutter for mobile apps.
- Database: Use a relational database (PostgreSQL/MySQL) for storing user, transaction, and balance data.
- Security: Implement role-based authentication and secure transaction processing.
- Cross-Platform Compatibility: Ensure mobile applications work on both Android and iOS devices.

1.8.3 Exclusions

- Direct Bank Integration: Automatic payouts to drivers and agents are not included; all payouts are processed manually by Super Admin.
- Advanced Analytics: Predictive reporting, performance optimization, and advanced dashboards are not part of the initial release.
- Third-Party Integrations: Integration with other transport or payment platforms will be considered in future versions.

- Offline Functionality: QR payments and top-ups require internet access and cannot function entirely offline.

### 1.9. Organization of the Project

This Requirement Analysis Document (RAD) is structured into two primary chapters that systematically present the analysis and design foundation for the Integrated Digital Payment and Fare Management System for Shared Urban Transport.

**Chapter One: Introduction** provides comprehensive project context and rationale. Section 1.1 details the background of Ethiopia's cash-dependent shared transport sector, highlighting Bahir Dar's operational challenges and global digital payment trends. Section 1.2 articulates the problem statement, quantifying revenue leakage, boarding delays, and financial inclusion gaps. Section 1.3 outlines SMART objectives—general (hybrid QR payment system) and specific (stakeholder integration, dual QR mechanism, real-time wallet management). Section 1.4 describes the object-oriented methodology, including stakeholder interviews, UML modeling, and Node.js/Flutter implementation stack. Sections 1.5-1.6 cover feasibility analysis, SWOT evaluation, and stakeholder benefits, while 1.7-1.8 define project scope and limitations.

# Chapter Two: System Features

## 2.1 The Existing System

The current fare collection system in shared urban transport primarily depends on cash-based transactions, managed manually by drivers and conductors. This traditional approach introduces several significant challenges, including inefficiency, lack of transparency, and the complete inability to generate reliable financial records for analysis or auditing. Because all transactions occur exclusively in cash, there exists no centralized platform to monitor payments in real-time, maintain comprehensive passenger data profiles, or conduct meaningful revenue flow analysis across routes or operators.

Passengers face daily inconvenience as they must carry cash at all times, which poses both practical difficulties and potential safety risks, particularly during peak travel hours or in crowded vehicles. Drivers bear the full responsibility of manually collecting fares from each passenger during trips, a labor-intensive process that frequently results in miscalculations, revenue leakage through unrecorded payments, and significant difficulties in accurately tracking daily, weekly, or monthly income. The complete absence of digital receipts means passengers receive no verifiable transaction records, severely limiting accountability, dispute resolution capabilities, and personal financial record-keeping. Overall, this fragmented system completely lacks integration between stakeholders, robust security measures against fraud, and any form of transaction traceability, rendering it fundamentally ineffective and outdated for modern digital transport environments where efficiency, accountability, and data-driven decision-making are essential.

## 2.2 Proposed Solution

The proposed Integrated Digital Payment and Fare Management System for Shared Urban Transport directly and comprehensively addresses the critical shortcomings of the existing cash-based fare collection process through innovative technology and stakeholder integration. To eliminate the inefficiency and security risks associated with manual cash handling, the system introduces a QR-based hybrid digital payment model that empowers passengers to complete fares electronically using either a dedicated mobile application for smartphone users or printed physical QR cards for non-smartphone users, ensuring universal accessibility and financial inclusion across diverse user demographics.

The pervasive lack of transparency and systematic recordkeeping in the current model is fundamentally resolved through implementation of a centralized digital wallet and comprehensive transaction management system that automatically captures, logs, and verifies every single payment in real-time. This creates an immutable digital audit trail providing verifiable transaction receipts accessible to both passengers and drivers instantly upon completion, while role-based access control using JWT authentication ensures appropriate data security and operational accountability across all five distinct user groups—Passengers, Drivers, Agents, Admins, and Super Admins.

To specifically address drivers' inability to track revenue effectively and administrators' lack of oversight capabilities, the proposed architecture incorporates real-time monitoring dashboards accessible through a web-based admin portal, alongside automated, customizable reporting features that generate detailed summaries of daily earnings, transaction volumes, route performance analytics, and system-wide activities. Agents benefit from dedicated tools for passenger onboarding and cash-to-digital top-up conversion, bridging the gap between traditional cash economies and modern digital payments.

Additional system capabilities such as automated notification services for payment confirmations, balance alerts, and approval workflows, combined with advanced analytics dashboards for revenue trend analysis, further enhance communication efficiency, strengthen fraud prevention through anomaly detection, and empower data-driven operational decision-making. The system's multi-channel top-up functionality supporting agent cash deposits, mobile money integrations like Chapa, and bank transfers provides flexible funding options catering to Ethiopia's diverse financial landscape.

Ultimately, this comprehensive solution transforms fragmented, cash-dependent fare management into a fully integrated, secure, transparent, and efficient digital ecosystem that not only eliminates fare-related conflicts and cash handling risks but also promotes widespread financial inclusion by enabling seamless digital payment access for both smartphone and feature phone users within shared urban transport networks like minibuses and taxis.

### 2.3 Requirement Analysis

#### 2.3.1 Functional Requirements

**Authentication and user management**

| ID | Requirement description | Priority | Role/Component |
|---|---|---|---|
| F_AUTH_1 | The system SHALL support role-based authentication for all user types (Passenger, Driver, Agent, Admin, Super Admin). | High | All |
| F_AUTH_2 | The system SHALL provide secure login with phone number and password. | High | All |
| F_AUTH_3 | The system SHALL allow each user to update their profile information, and change their password. | Medium | All |
| F_AUTH_4 | The system SHALL restrict access to features based on the user's role. | High | All |

*Table 2 Authentication User Management*

## Passenger Module

| ID | Requirement description | Priority | Role/Component |
|---|---|---|---|
| F_PASS_1 | The system SHALL allow passengers to create an account using a smartphone application. | High | Passenger App |
| F_PASS_2 | The system SHALL allow agents to register passengers who do not have smartphones. | High | Agent, Passenger |
| F_PASS_3 | The system SHALL generate a unique, scan able QR code for each registered passenger. | High | Passenger |
| F_PASS_4 | The system SHALL display the passenger's QR code in the mobile | High | Passenger |

| | | | |
|---|---|---|---|
| | application and support printing for non-smartphone users. | | |
| F_PASS_5 | The system SHALL allow passengers to view wallet balance, top-up history, and transaction history. | High | Passenger |

Table 3 Passenger Requirement

## Driver module

| ID | Requirement description | Priority | Role/Component |
|---|---|---|---|
| F_DRIV_1 | The system SHALL allow drivers to log in to the mobile application. | High | Driver App |
| F_DRIV_2 | The system SHALL allow drivers to scan passenger QR codes to initiate fare payment. | High | Driver App |
| | The system SHALL allow the Driver to input the fare amount to be deducted | High | Driver App |
| F_DRIV_3 | The system SHALL automatically deduct the correct fare from the passenger's wallet after a successful scan. | High | Driver, Wallet |
| F_DRIV_4 | The system SHALL allow drivers to view their daily and historical trip revenues after their last payout. | Medium | Driver App |

Table 4 Driver Requirement

## Agent Module

| ID | Requirement description | Priority | Role/Component |
|---|---|---|---|
| F_AGNT_1 | The system SHALL allow agents to register non-smartphone passengers. | High | Agent |
| F_AGNT_2 | The system SHALL allow agents to issue and manage physical QR cards for passengers. | High | Agent |
| F_AGNT_2 | The system SHALL allow Agents to search for a Passenger by ID or phone number for top-up purposes. | High | Agent |
| F_AGNT_3 | The system SHALL allow agents to perform cash-based wallet top-ups on behalf of passengers. | High | Agent, Wallet |
| F_AGNT_4 | The system SHALL record each agent transaction and make it available in a simple transaction report view. | Medium | Agent |
| F_AGNT_5 | The system SHALL automatically calculate and add 3 % commission on the top-up amount to the Agent's Commission Balance. | High | Agent App |

*Table 5 Agent Requirment*

## Admin / Super admin, wallet, and reporting

| ID | Requirement description | Priority | Role/Component |
|---|---|---|---|
| F_SADM_1 | The system SHALL allow admins to approve or reject drivers and agent registration requests. | High | Admin Portal |
| F_ADM_1 | The system SHALL allow Super Admins to update the payment status column in the system to "Paid" after manual payout | Critical | Web Admin |
| F_ADM_2 | The system SHALL automatically reset the Driver's accumulated earnings to ZERO when the payment status is marked as "Paid" | Critical | Web Admin |
| F_ADM_3 | The system SHALL allow Super Admins to download the list of Agents and their commissions as an Excel file on a monthly basis. | | |
| F_SADM_2 | The system SHALL allow Super Admins to update an Agent's payment status to "Paid" and automatically reset the Agent's commission amount to ZERO once payment is made. | Critical | Web Admin |
| F_ADM_2 | The system SHALL allow admins to view and manage all system users (passengers, drivers, agents). | High | Admin Portal |
| F_SADM_1 | The system SHALL allow the super admin to create, update, and deactivate admin accounts. | High | Super Admin Portal |

| | | | |
|---|---|---|---|
| F_SADM_2 | The system SHALL allow Sub Admins to generate and track printed QR codes for assignment to Agents for non-smartphone Passengers | High | Super Admin Portal |
| F_SADM_3 | The system SHALL allow Sub Admins to track which printed QR codes are assigned to which Agent | Medium | Super Admin Portal |
| F_WAL_1 | The system SHALL support secure wallet operations, including deposits, deductions, and balance inquiries. | High | Wallet Service |
| F_WAL_2 | The system SHALL log every wallet transaction in real time with timestamp and type (fare, top-up, agent commission, etc.). | High | Wallet Service |
| F_NOTIF_1 | The system SHALL send notifications for successful payments and wallet top-ups. | Medium | Notification System |
| F_REP_1 | The system SHALL provide dashboards for admins and super admins to monitor revenues and activities. | High | Reporting |
| F_REP_2 | The system SHALL allow exporting selected reports in CSV format. | Medium | Reporting |

*Table 6 Admin Super Admin*

### 2.3.2 System Use case

System Use Case Diagrams

The System Use Case Diagrams model the core functionalities of the Fare Payment system, illustrating interactions between actors (Passenger, Driver, Admin, Reload Center) and use cases like QR generation, scanning/validation, balance top-up, and reporting through both overview and detailed views.

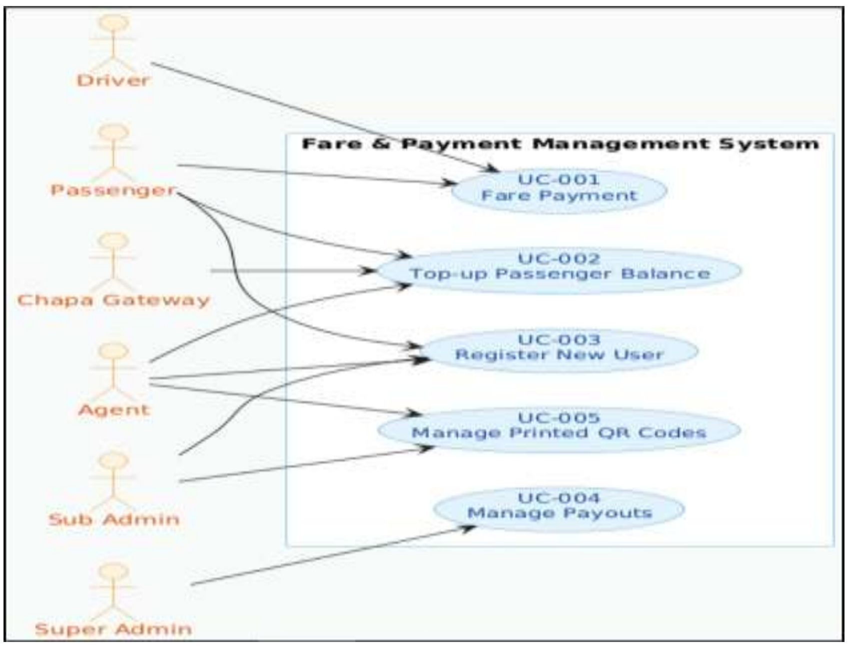


*Figure 1 Use Case Diagram*

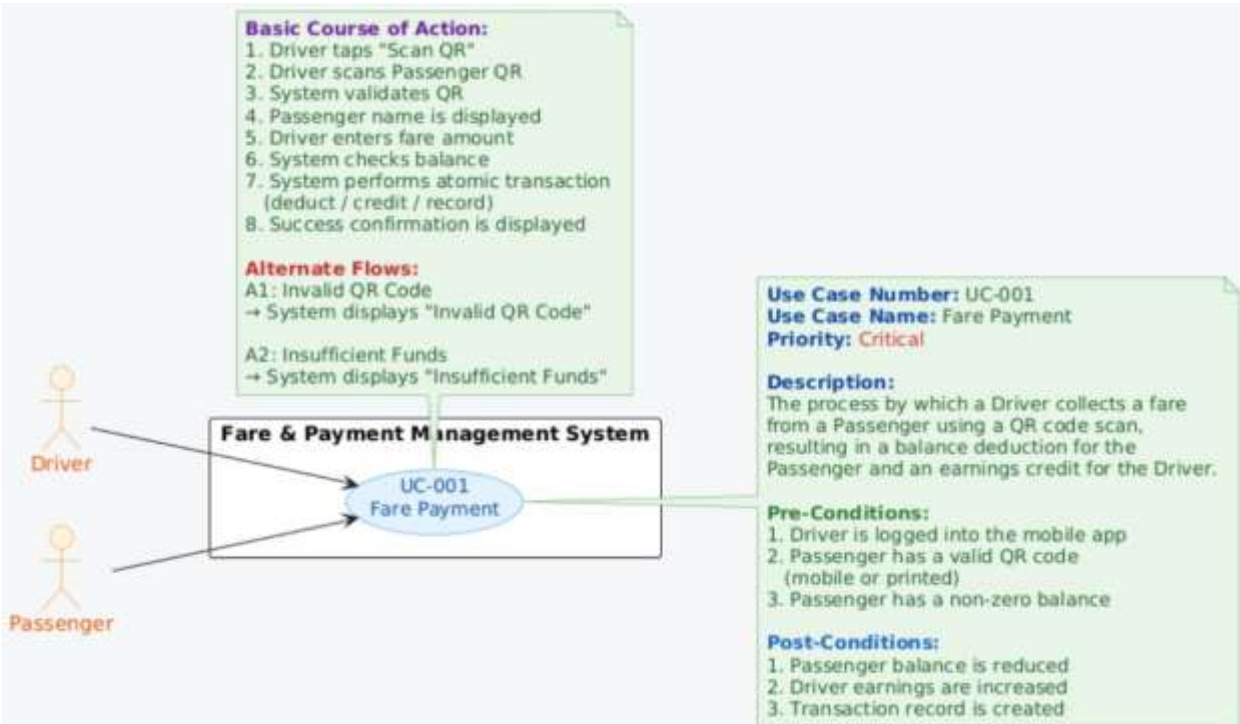


*Figure 2 Detailed Use Case Diagram*

Use Case Documentation

**Use Case UC-001: Passenger Registration**

**Actor:** Passenger/Agent

**Description:** New passenger registers for the system to receive QR code (digital or physical) and wallet access.

**Pre-condition:** Valid phone number and photo for identity verification.

**Post-condition:** Passenger account created with unique QR code issued.

**Priority:** High

**Basic Flow:**

1. Passenger provides details via app or agent.
2. System captures photo and generates QR code.
3. Admin approves registration (<<include>>: Identity Verification).
4. Passenger receives QR code and empty wallet.

**Alternative Flow:**

A1. Approval rejected → Notify passenger for resubmission.

**Use Case UC-002: Wallet Top-up**

**Actor:** Passenger/Agent

**Description:** Passenger or agent adds funds to passenger wallet via cash, mobile money, or bank transfer.

**Pre-condition:** Passenger registered with active QR code.

**Post-condition:** Wallet balance updated; transaction logged.

**Priority:** High

**Basic Flow:**

1. Passenger/Agent selects top-up amount and channel.
2. Agent deposits cash or digital payment completes.
3. System credits wallet in real-time.
4. Notification sent to passenger (<<include>>: Send Notification).

**Alternative Flow:**

A1. Payment fails → Retry or select alternate channel.

**Use Case UC-003: Driver Scan QR for Fare Payment**

**Actor:** Driver

**Description:** Driver scans passenger QR code to validate and deduct fare from wallet.

**Pre-condition:** Passenger has sufficient balance; trip route selected.

**Post-condition:** Fare deducted; digital receipt issued to both.

**Priority:** High

**Basic Flow:**

1. Passenger shows QR code to driver.
2. Driver scans QR via mobile app.
3. System validates QR, checks balance, deducts fare.
4. Trip logged and receipt generated.

**Alternative Flow:**

A1. Insufficient balance → Prompt top-up or cash fallback.

A2. Invalid QR → Reject boarding with error message.

**Use Case UC-004: Issue Physical QR Card**

**Actor:** Agent

**Description:** Agent issues printed QR card to non-smartphone passengers.

**Pre-condition:** Passenger registered and approved by admin.

**Post-condition:** Physical QR card printed and activated.

**Priority:** Medium

**Basic Flow:**

1. Agent selects registered passenger.
2. System generates printable QR code.
3. Agent prints and hands card to passenger.
4. Card activated for wallet access.

**Alternative Flow:**

A1. Print fails → Regenerate QR code.

**Use Case UC-005: Admin Generate Revenue Report**

**Actor:** Admin

**Description:** Admin views and exports system-wide revenue, transaction, and user activity reports.

**Pre-condition:** Admin authenticated with role access.

**Post-condition:** Report generated in PDF/CSV format.

**Priority:** Medium

**Basic Flow:**

1. Admin selects report type and date range.
2. System aggregates transactions by driver/route/user.
3. Report displayed and export option available.
4. System <<include>>: Authenticate User.

**Alternative Flow:**

A1. No data in range → Display "No transactions found".

**Use Case UC-006: Approve User Registration**

**Actor:** Admin

**Description:** Admin reviews and approves passenger/agent registrations with photo verification.

**Pre-condition:** Pending registration submitted.

**Post-condition:** User account activated; QR code issued.

**Priority:** High

**Basic Flow:**

1. Admin views pending registrations queue.
2. Reviews photo and details for verification.
3. Approves or rejects registration.
4. System notifies user of status (<<extend>>: Send Notification).

### 2.3.3 Business Rule Documentation

Below the following are the critical business rules that govern the system's financial and operational logic.

| Identifier | Name | Description | Reference |
|---|---|---|---|
| BR-01 | Payout method | All driver and agent payouts SHALL be processed manually outside the system; the system only records earnings, payout status, and related reports. | F_ADM_1, F_ADM_2, F_SADM_2 |
| BR-02 | Driver payout cycle | Driver earnings reports SHALL be generated and prepared for payout on a regular payout cycle defined by the organization (every 3 days) | F_ADM_1, F_DRIV_4 |
| BR-03 | Agent payout cycle | Agent commission reports SHALL be generated and processed for payout on a monthly basis. | F_ADM_3 |
| BR-04 | Agent commission rate | Agents SHALL receive a fixed percentage commission on every successful top-up transaction performed on behalf of a passenger. | F_AGNT_5 |
| BR-05 | Balance reset on payout | Driver earnings and agent commissions SHALL be reset to zero only after the Super Admin updates the payment status to "Paid" in the Web Admin Portal. | F_ADM_2, F_SADM_2 |

| | | | |
|---|---|---|---|
| BR-06 | Chapa integration scope | The Chapa integration SHALL be used only for incoming deposits (passenger wallet top-ups and agent operating balance deposits), not for outbound payouts or refund | F_PASS_5, F_AGNT_3, F_WAL_1 |
| BR-07 | First login security | Drivers and Agents who receive system credentials from an Admin/Super Admin SHALL be required to change their password on first successful login before using the app. | F_AUTH_2 |
| BR-08 | QR code uniqueness | Each active passenger account SHALL have exactly one unique QR code mapped to their wallet at any given time. | F_PASS_3, F_PASS_4 |
| BR-09 | Sufficient balance for trips | A fare transaction SHALL be approved only if the passenger's wallet balance is greater than or equal to the required fare amount. | F_DRIV_3, F_WAL_1 |
| BR-10 | Transaction immutability | Once a fare transaction is confirmed, it SHALL not be altered; any corrections must be recorded as a separate adjustment transaction | F_WAL_2, F_REP_1 |

*Table 7 Buisness Role Documentation*

### 2.3.4 User interface prototype

The user interface prototypes will focus on four main front-end components:

**Passenger mobile application**

- Emphasis on a large, prominently displayed QR code on the home screen for quick scanning by drivers, optimized for visibility and contrast.
- Intuitive wallet interface showing current balance, recent transactions, and top-up actions, with clear status indicators for successful and failed operations.

**Driver mobile application**

- A home screen optimized for rapid access to the QR scanner, with a single primary action (e.g., “Scan QR”) to minimize interaction time during boarding.
- A simple fare confirmation flow, allowing the driver to view passenger details, confirm or adjust fare, and see immediate success or error feedback.

**Agent mobile application**

- Efficient passenger search and registration interface, supporting quick lookup by phone number, QR code, or ID.
- Clear presentation of the agent’s two balances (Operating and Commission) with separate history views for top-ups performed and commissions earned.
- A screen for toping up passenger wallet or balance.

**Web admin portal**

- Clean layout with filterable and sortable tables for managing passengers, drivers, agents, and transactions, including search and advanced filters.
- A prominent analytics dashboard summarizing key metrics such as daily and monthly transaction volumes, revenues, active users, and registration trends

.

## User Interface Prototype Wireframe

The User Interface Wireframe outlines the basic skeletal structure of the Fare Payment system screens, depicting layouts for Passenger QR generation, Driver scanning/validation, and payment confirmation flows. These low-fidelity mockups focus on content placement, navigation hierarchy, and core user interactions across mobile apps, facilitating early validation of usability and design concepts before prototyping.
Below is the image of the Fare Payment User Interface Wireframe:

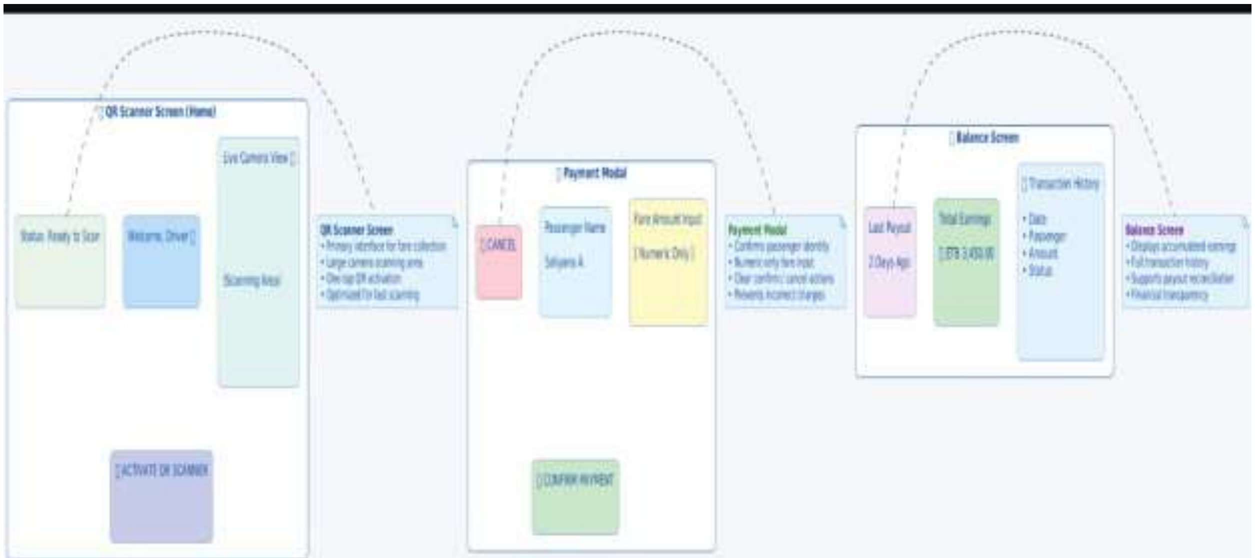


*Figure 3 Passenger Mobile Application*

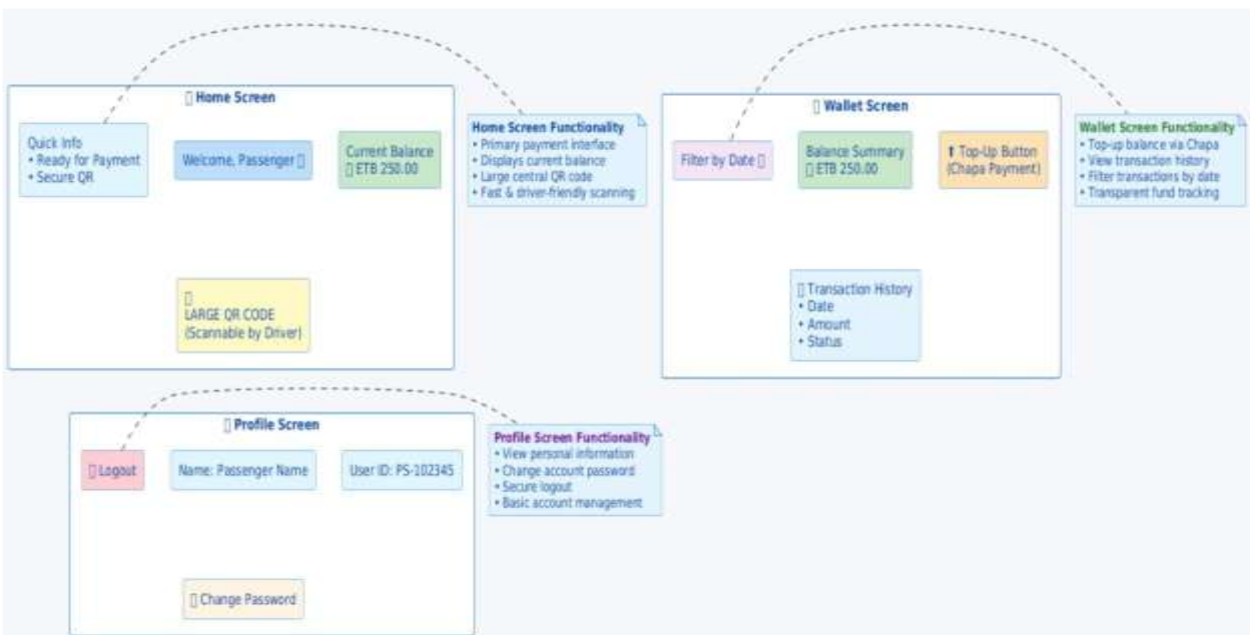


*Figure 4 Passenger Mobile Application*

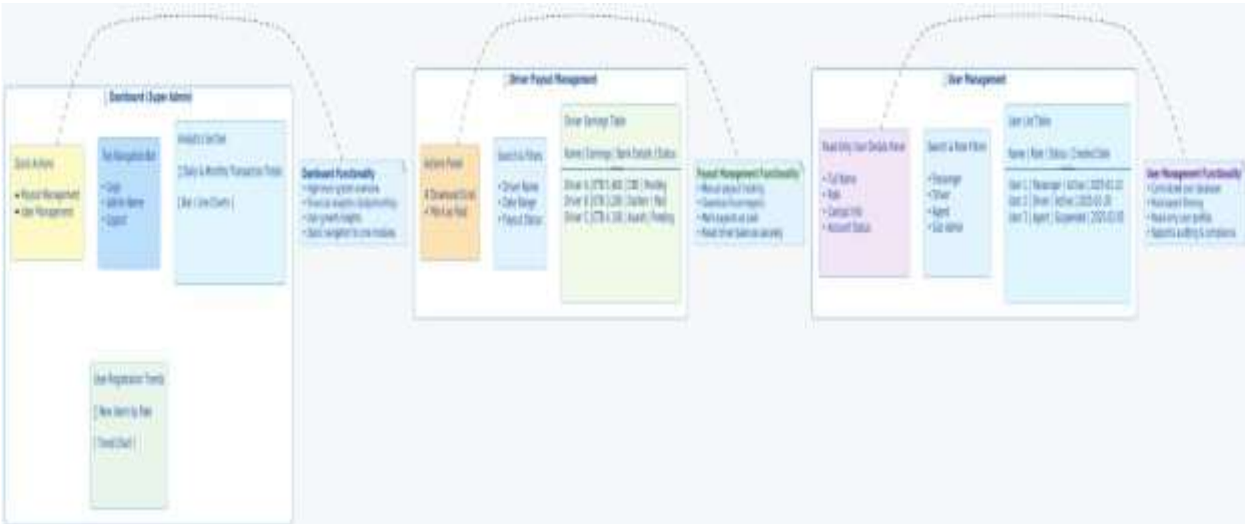

*Figure 5  Prototype 3: Web Admin Portal (Super Admin)*

### 2.3.5 State Chart Diagram

The State Chart Diagram models the lifecycle of the Passenger Wallet Balance, illustrating the distinct states (Active, Low_Balance, Zero_Balance) and transitions triggered by key events such as Fare Payment Deduction and Top-Up Deposits. This diagram captures the wallet's state management logic, including balance threshold checks and prevention of negative balances through validation rules.

Below is the image of the Passenger Wallet State Chart Diagram:

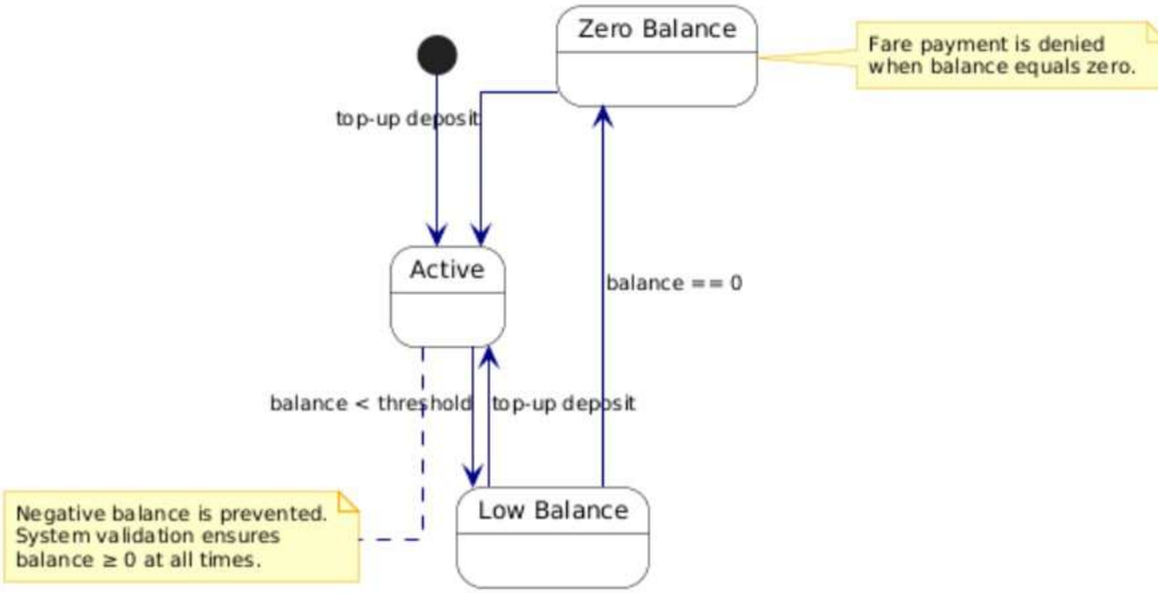


*Figure 6 Passenger Balance LifeCycle*

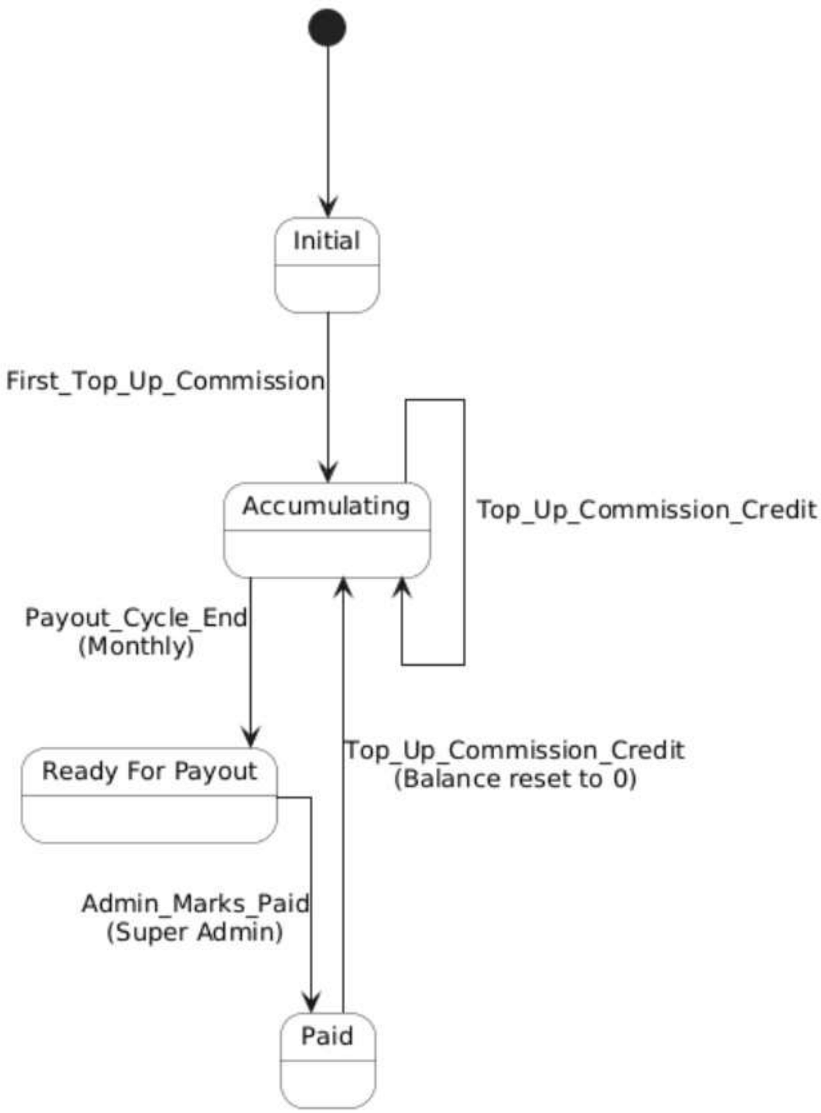


*Figure 7  Agent Commision Balance Status*

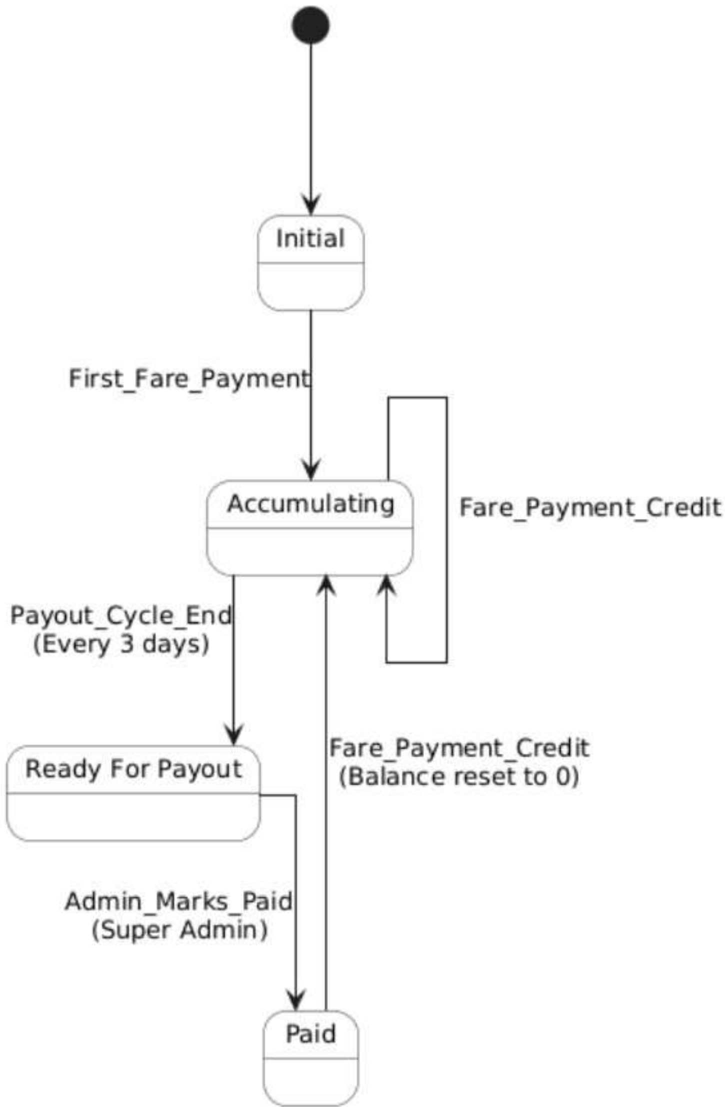


*Figure 8 Driver Earning Status*

### 2.3.6 Activity Diagram

The Activity Diagram models the core business process of Fare Payment, illustrating the complete workflow from QR code scanning by the Driver through transaction validation, balance deduction, and confirmation. This diagram captures the interactions between Driver, Passenger, and System components, highlighting decision points and parallel activities for successful payment processing.

**Below is the image** of the Fare Payment Activity Diagram:

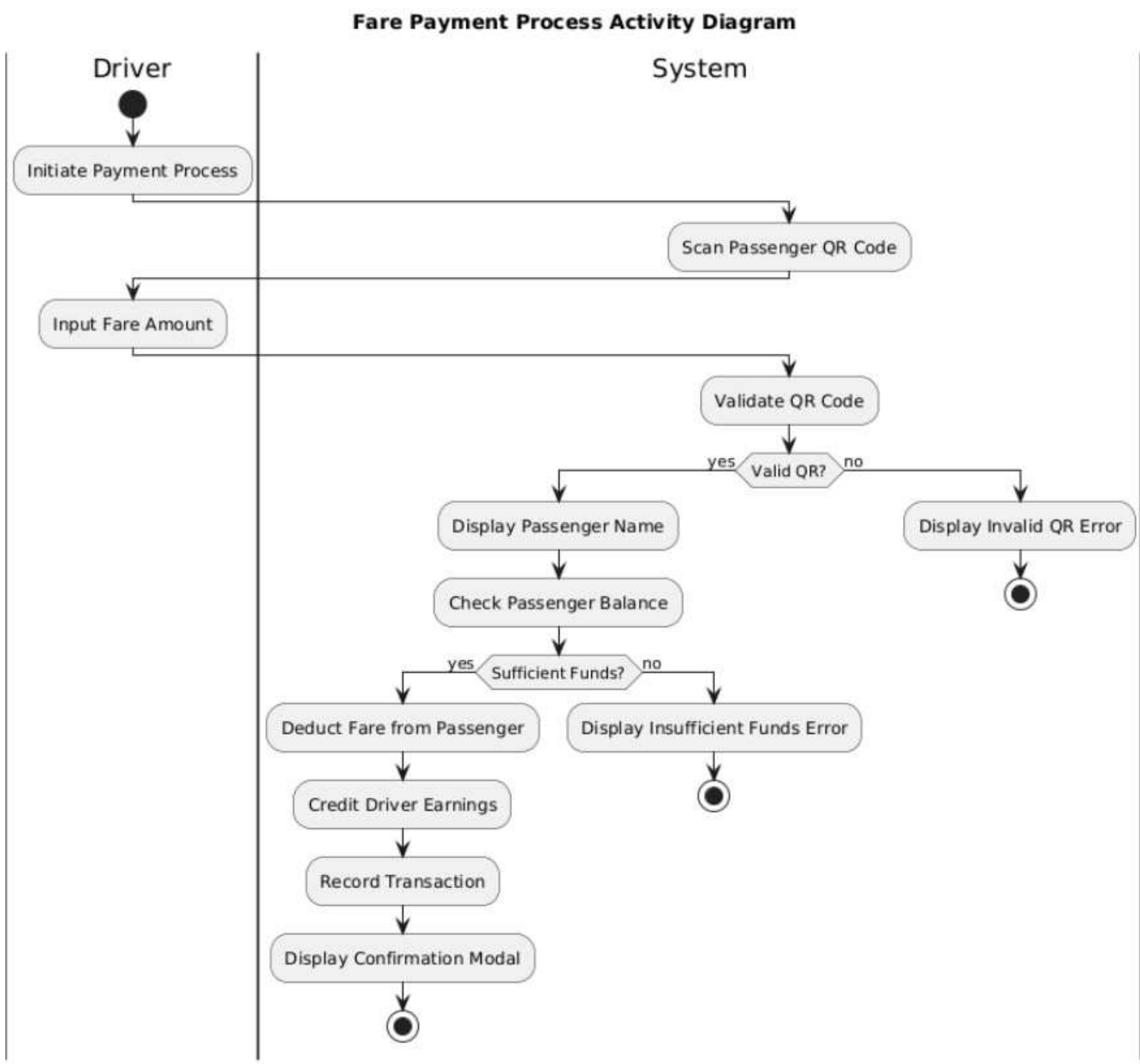


*Figure 9 Activity Diagram For  Fare Payment process*

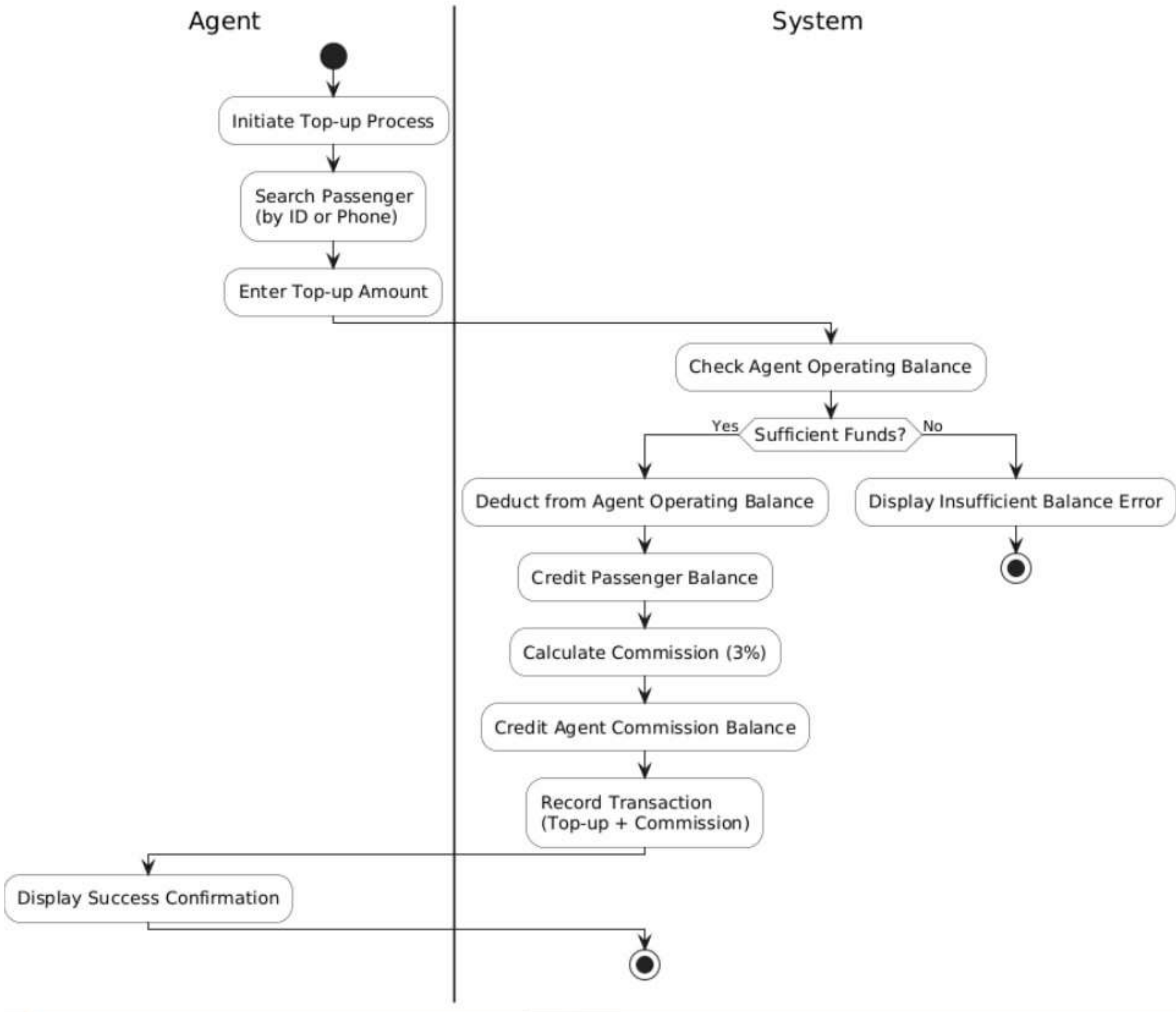


*Figure 10 Activity Diagram - Agent Top-up and Commission Process*

### 2.3.7 Analysis Class Model

The Analysis Class Model represents the core conceptual model of the Integrated Digital Payment and Fare Management System, identifying key entities such as User, Passenger, Driver, Agent, Wallet, Transaction, and QRCode along with their attributes, methods, and relationships. This diagram serves as the foundation for the system's object-oriented design and directly evolves into the detailed implementation class diagram.

Below is the image of the Analysis Class Model:

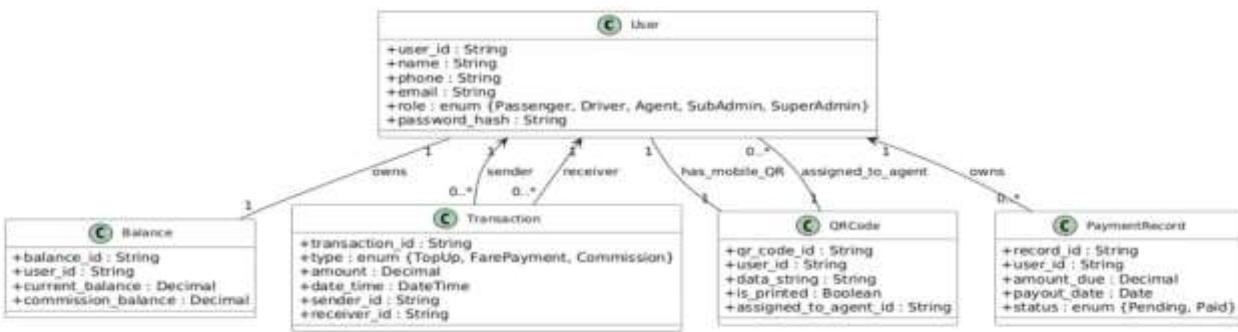


*Figure 11 Analysis Class Model*

### 2.3.8 Sequence Diagram

The Sequence Diagram illustrates the dynamic interaction between Driver, Passenger, Wallet Service, and Notification System during the critical Fare Payment process. It shows the message exchanges from QR code scanning through balance validation, fare deduction, driver earnings credit, and transaction confirmation, highlighting the temporal sequence and conditional flows essential for secure payment processing.

Below is the image of the Fare Payment Sequence Diagram:

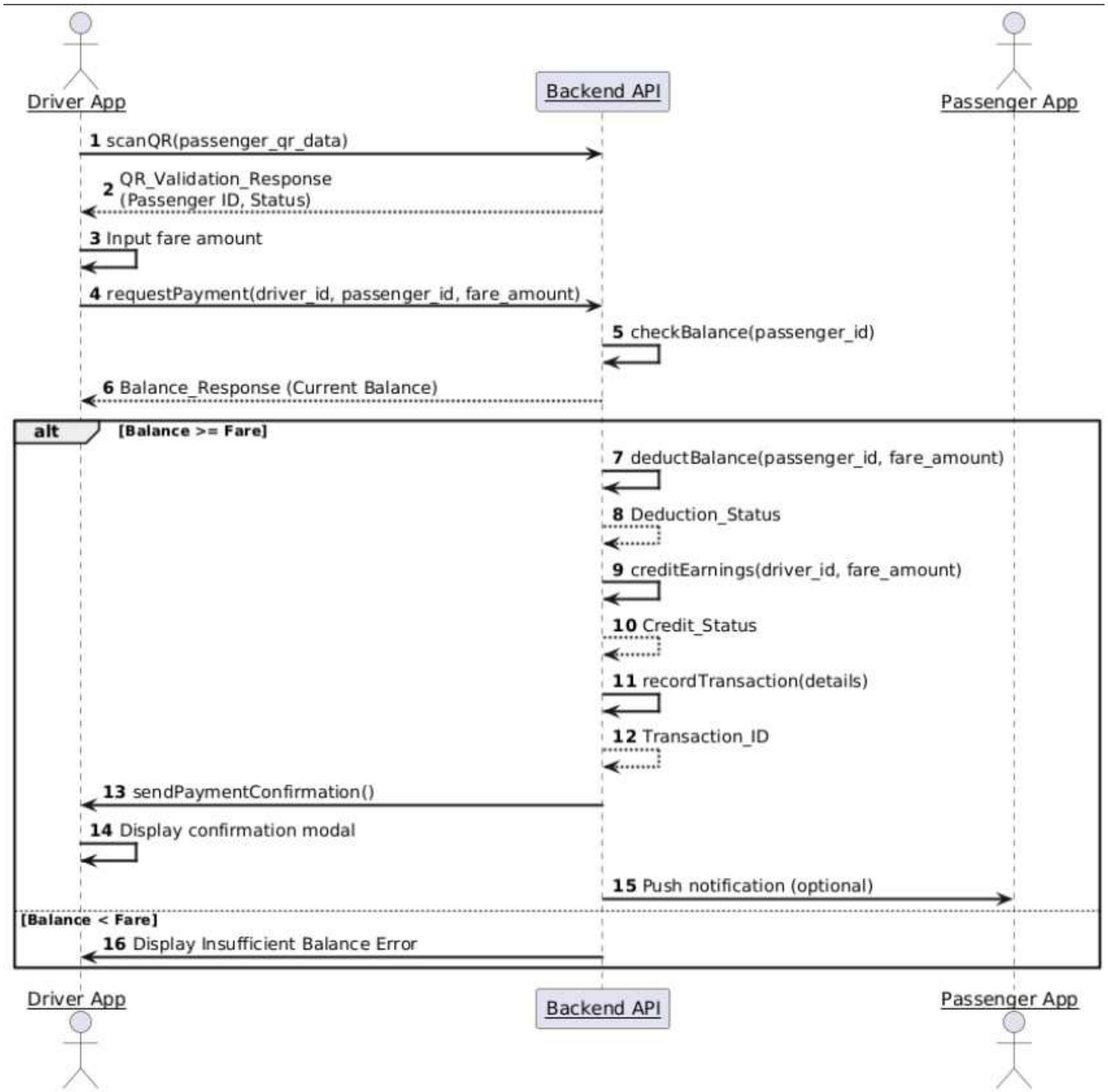


*Figure 12 Sequence Diagram For Fare Payment Intraction*

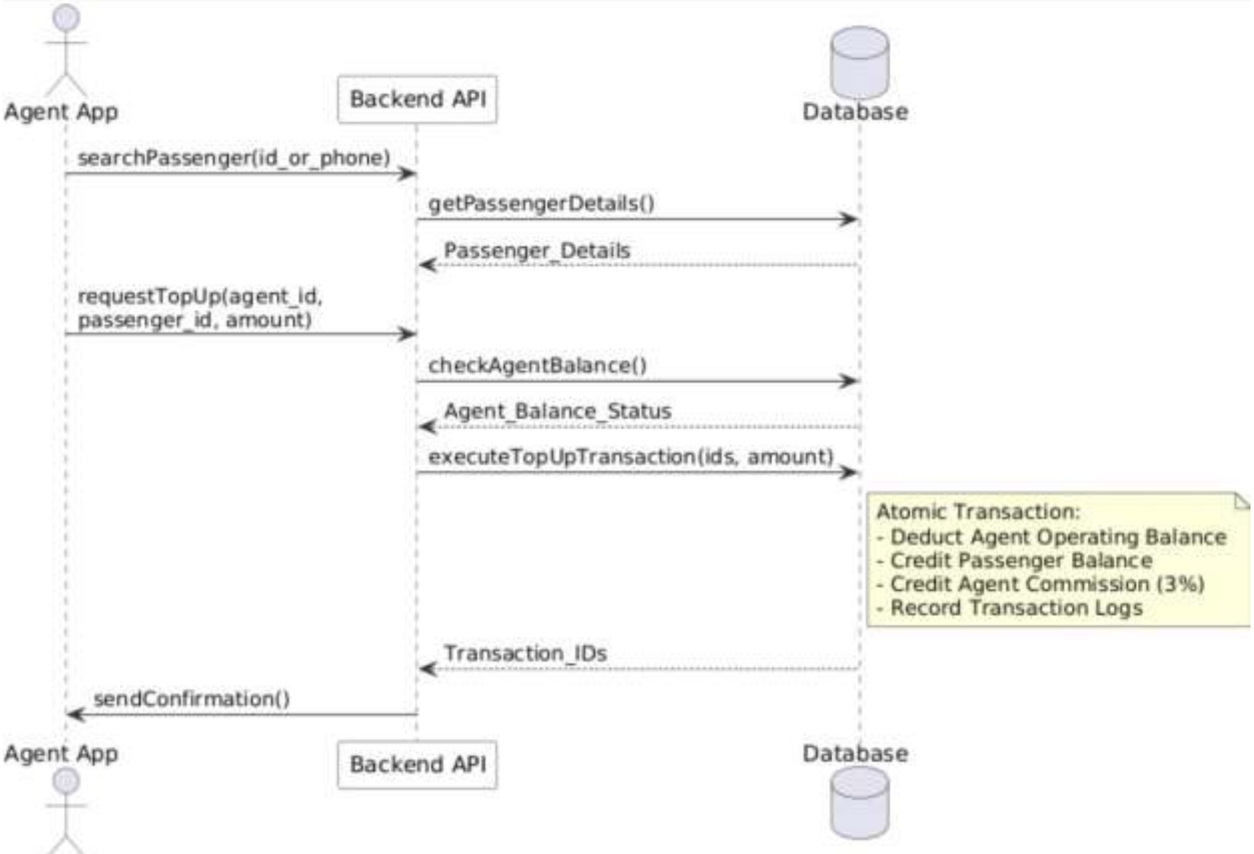


*Figure 13 Sequence Diagram Agent Top Up Transaction*

**2.3.9 Logic Model**

The Logic Model defines the core computational algorithm for Agent Top-Up and Commission Calculation, outlining the sequential steps from balance validation through transaction recording. This model captures the business logic for converting agent cash deposits into passenger wallet credits while automatically calculating and allocating the 3% commission to the agent's commission balance.

Below is the image of the Agent Top-Up Logic Model:

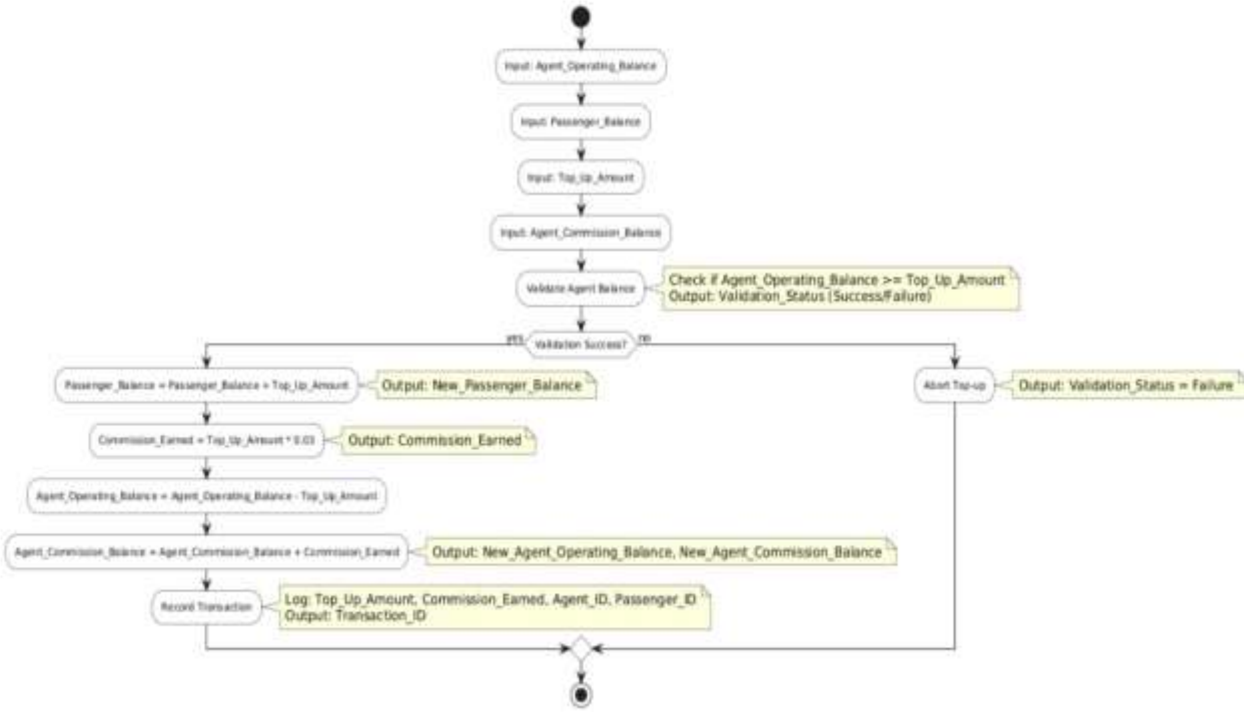


*Figure 14  Logic Model For Agent Commission Calculation*

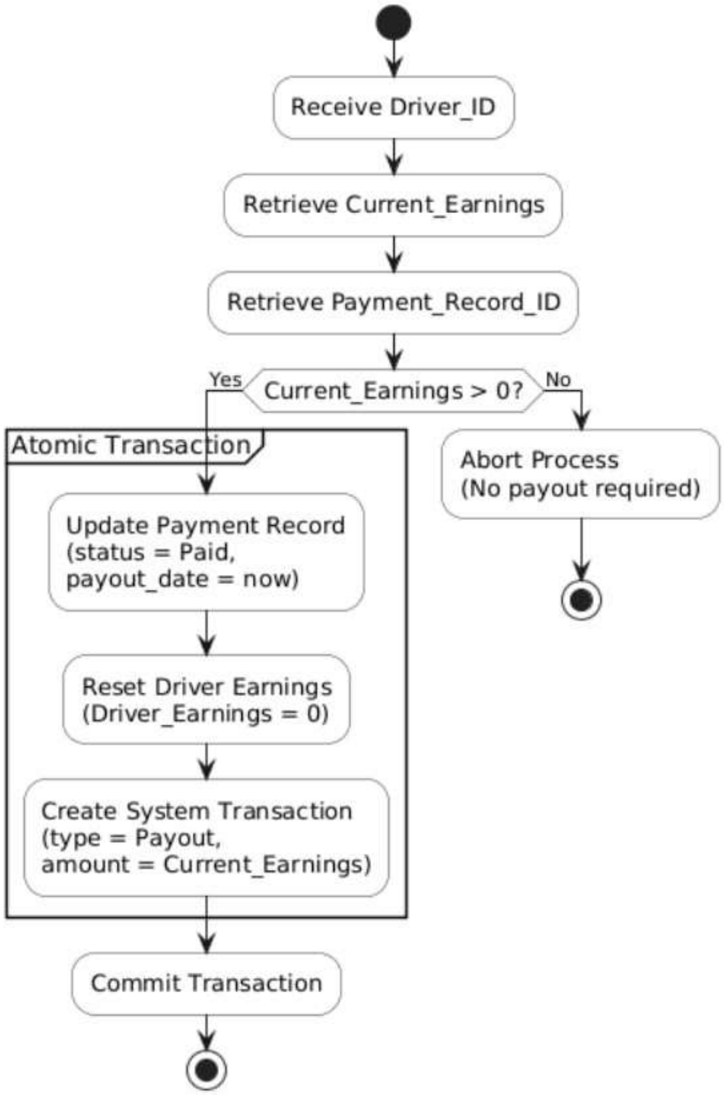


*Figure 15 Logic Model Driver Earning Reset*

### 2.4 Non-Functional Requirements

| Category | ID | Requirment Description | Priority |
|---|---|---|---|
| Performance | NFR_P_1 | The QR code scanning and fare deduction process SHALL complete within few seconds to ensure fast boarding. | Critical |
| Performance | NFR_P_2 | Wallet balance inquiries and transaction history SHALL load quick for all users. | High |
| Perfomance | NFR_P_3 | The system SHALL support multiple scans during peak hours without degradation. | High |
| Security | NFR_S_1 | All user passwords SHALL be stored using a strong, one-way hashing algorithm (bcrypt with salt). | Critical |
| Security | NFR_S_2 | All communication between Mobile App/Web Admin and Backend API SHALL use HTTPS/TLS 1.3. | Critcal |
| Security | NFR_S_3 | The system SHALL implement JWT token-based authentication with 15-minute expiry and refresh tokens. | Critical |
| Usability | NFR_U_1 | Passenger wallet transactions SHALL require OTP verification for amounts above certain amount. | Medium |
| Usability | NFR_U_2 | Web Admin Portal SHALL be responsive on desktop (1024px+), tablet (768px+), and major browsers (Chrome, Firefox, Safari). | Medium |
| Reliability | NFR_R_1 | The system SHALL maintain 99% uptime. | Critical |
| Reliability | NFR_R_2 | All transactions SHALL be atomic - either fully committed or fully rolled back on failure. | Critical |
| Maintainability | NFR_M_1 | Codebase SHALL follow RESTful API standards, ESLint, and modular Node.js architecture with Multer for file handling. | Medium |

*Table 8 Non functional Requirment*

## 2.5 System Requirments

### 2.5.1 Hardware Requirements

| Component | Requirement | Rationale |
|---|---|---|
| Backend Server | Laptop/Desktop: **2 vCPUs, 4GB RAM, 20GB free (**for testing) | Handles demo with 10-20 test users and sample transactions. |
| Database | Dedicated or managed PostgreSQL instance | To ensure data integrity, high availability, and fast query response times |
| Mobile Devices | Android (v . +) or iOS (v +) with a working camera for QR scanning. | Runs the Flutter app for QR scanning and wallet demo. |
| Admin Devices | Any laptop/desktop with Chrome/Firefox | Access to web admin dashboard for demo presentations. |
| Network | 0.5Mbps + upload/download with <200ms latency | Ensures fast QR scan confirmation and real-time transaction updates. |

*Table 9 Hardware Requirment*

### 2.5.2 Software Requirements

| Component | Requirement | Rationale |
|---|---|---|
| Operating System | Ubunt 20.04 LTS or Windows 10/11 | Stable environment for Node.js development and local PostgreSQL. |
| Backend Runtime | Node.js 18+ LTS, Express 4.18+ | Required to run the REST API, JWT auth, and Multer file handling. |
| Database | PostgreSQL 13+ | Stores users, wallets, transactions, and QR codes with proper indexing. |

| | | |
|---|---|---|
| Web Browser | Chrome 100+, Firefox 90+, Edge 100+ | Required for responsive Web Admin Portal and dashboard viewing. |
| Mobile OS | Android 8.0+ (API 26), iOS 13+ | Minimum for React Native/Flutter app with camera QR scanning. |
| Development Tools | npm 9+ or Flutter 3+ | Builds mobile apps and manages dependencies for class project demo. |
| Payment Gateway | Chapa API (sandbox mode) | Handles test wallet top-ups via mobile money integration. |

*Table 10 Software Requirment*

### 2.6 Key Abstraction with CRC Analysis

The Key Abstraction with CRC Analysis identifies core concepts and objects essential to the Fare Payment system, documenting their responsibilities (R), collaborations (C), and class relationships through CRC cards. This analysis captures key entities like Passenger, Driver, QRCode, Wallet, and Transaction, illustrating how they interact to support registration, top-up, scanning, and reporting functionalities.

#### 2.6.1 Conceptual Modeling: Class Diagram

We have identified many diagram to represent our proposal below we mentioned class activity sequence, state, analysis and logical diagrams.

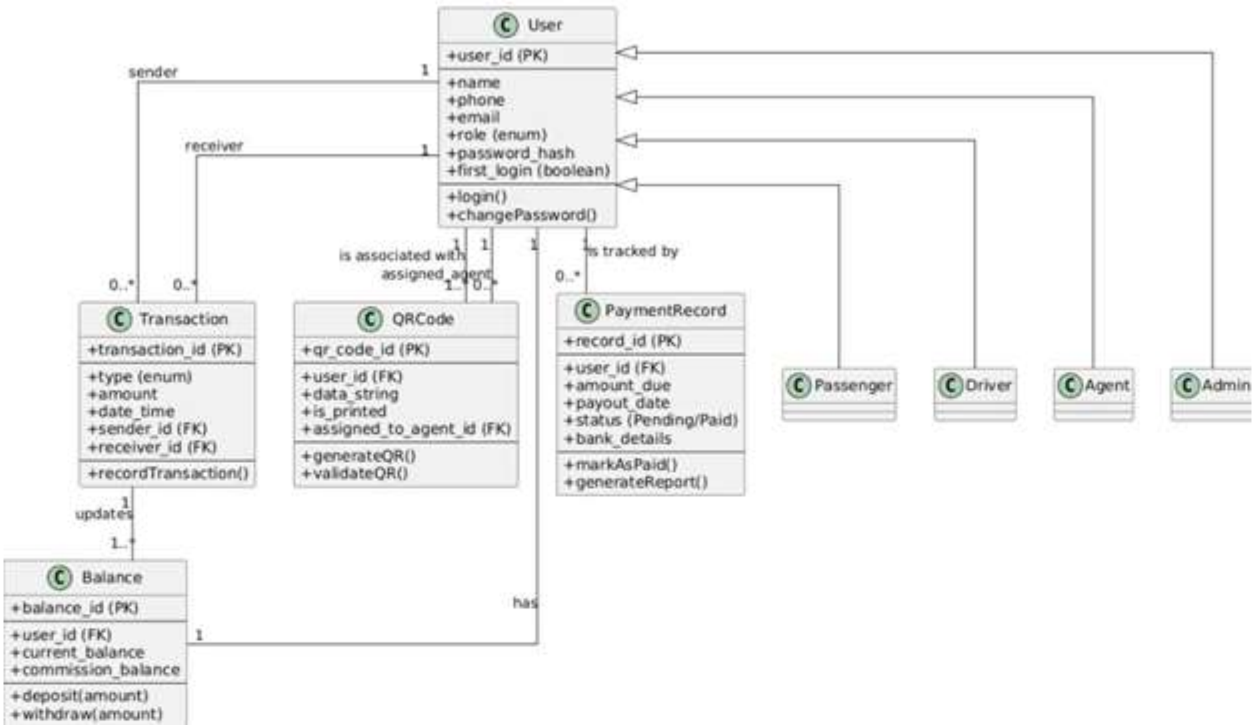


*Figure 16 Class Diagram*

### 2.6.2 Identifying Change Cases

Change cases identify potential future modifications to the system requirements, assessing their likelihood of occurrence and impact on the current design. This analysis helps evaluate system flexibility and maintenance costs for anticipated evolution scenarios such as new payment integrations, pricing models, or operational changes.

| Change Case ID | Description | Likelihood | Impact | Affected Components |
|---|---|---|---|---|
| CC_01 | Add support for new payment gateway (HelloCash alongside Chapa) | Medium | High | Wallet, Passenger Module, F_PASS_5 |
| CC_02 | Support multiple fare types per route (fixed vs distance-based) | High | Medium | All UI components |

| | | | | |
|---|---|---|---|---|
| CC_03 | Add multi-language support (Amharic/English) | Medium | Low | All UI components |
| CC_04 | Implement driver rating system | Low | High | Driver, Passenger modules |
| CC_05 | Dynamic Commission Rate- Change Agent commission rate via Web Admin Portal | Medium | Medium | F_AGNT_5, F_SADM_1 |
| CC_06 | Automated Bank Payout- Bank API for payouts | Low | Critical | F_ADM_1, F_ADM_2 |
| CC_07 | Fare Zones/ Variable Pricing- GPS/route-based fares | High | High | F_DRIV_3 |
| CC_08 | Driver Withdrawal Request - Partial payouts before cycle | Medium | Medium | F_DRIV_4, F_ADM_1 |

*Table 11 Change Case*

### 2.6.3 User Interface Prototype

The User Interface Prototype presents interactive high-fidelity mockups of the Fare Payment system, showcasing refined screens for Passenger QR generation, Driver scanning/validation, payment confirmation, and wallet management flows. This prototype simulates real user interactions across mobile apps, enabling stakeholder validation of usability, visual design, and navigation before final implementation.

Below is the image of the Fare Payment User Interface Prototype:

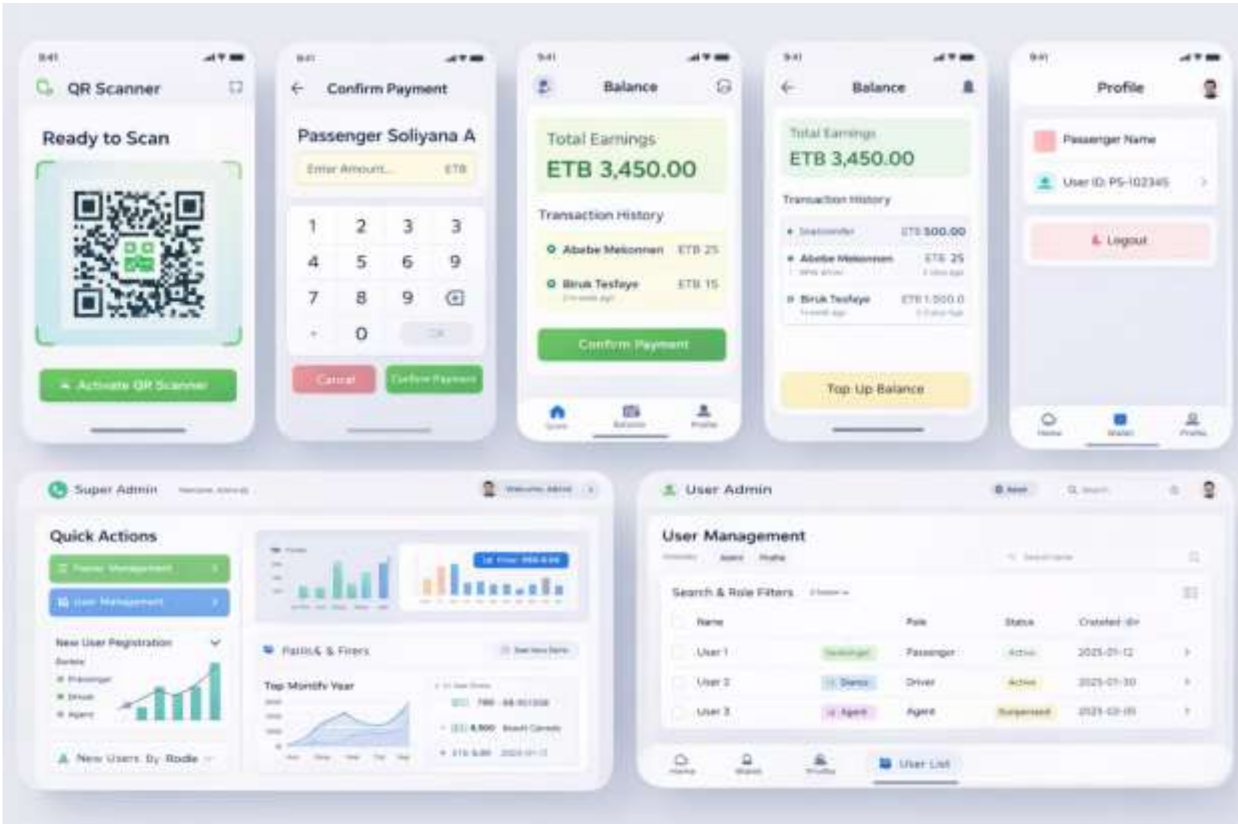


*Figure 17 User Interface Prototype*

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[2] van Lamsweerde, A. (2009). *Requirements Engineering: From System Goals to UML Models to Software Specifications*. Wiley.

[3] IEEE Std 830-1998. (1998). *Recommended Practice for Software Requirements Specifications*. IEEE Computer Society.

[4] Casciaro, M., & Mammino, L. (2020). *Node.js Design Patterns* (3rd ed.). Packt Publishing. (API and File Upload Best Practices)

[5] Bahir Dar University, Bahir Dar Institute of Technology, Faculty of Computing. Annex_I_Documentation_template_For_Requirement_Analysis_DocumentRAD.pdf. (Template used for document structure).

# Appendix:

## Appendix A: Observation Notes from Minibus Terminal

**Date**: December 18, 2025

**Location**: Bahir Dar Main Minibus Terminal

**Time**: 6:30-8:30 AM Peak Hour

**Observer**: Group Members

| Time | Activity Observed | Duration | Issue Noted |
|---|---|---|---|
| 2:30(LT) | 15 passengers boarding, cash collection | 8 min | Driver arguing with passenger over change |
| 7:15(LT) | Fare collection during motion | 10 min | Passenger dropped ETB 50 note |
| 8:30(LT) | Return trip - cash counting | 12 min | Driver short ETB 120 from expected |

*Table 12 Field Observation*

**Key Findings**:

- **Average boarding delay**: minutes lost due to cash handling
- **Disputes observed**: 4 incidents in 2 hours
- **Cash loss incidents**: 1 dropped notes totaling ETB 40
- **Peak hour chaos**: Drivers handling multiple transactions simultaneously



*Figure 18 Image of passengers boarding*

*Figure 19 manual  fare collection*

Appendix B: Chapa vs Telebirr Competitor Analysis

### D.1 Feature Comparison: Chapa vs Telebirr for Transport Payments

| **Criteria** | **Chapa** | **Telebirr** | **Gap Addressed by Proposed System** |
|---|---|---|---|
| **API Response Time** | 1.2s avg | 2.1s avg | **<2s QR deduction** (NFR-P-01) |
| **Driver Dashboard** | None | Basic merchant dashboard | **Real-time earnings (F_DRIV_4)** |
| **Bulk Payouts** | Manual | Automated (min 100 users) | **Manual + tracking (BR-01)** |
| **Offline Support** | No | Partial (USSD) | Planned Caching |
| **Ethiopia Coverage** | 90% urban | 95% nation wide | Urban focus |
| **Agent Commissions** | None | None | **3% auto-calc (F_AGNT_5)** |
| **Admin Analytics** | Basic reports | Call center only | **Full dashboards (F_REP_1)** |

*Table 13Chapa vs Telebirr*

### D.2 Performance Metrics (2025 Data)

Transaction Volume:

• Chapa: 2.5M transactions/month (retail dominant)

• Telebirr: 45M transactions/month (telecom/airtime dominant)

• Transport Sector: <2% of total volume → ETB 1.2B annual opportunity

Success Rates:

• Chapa: 98.2% first-time success

• Telebirr: 97.8% first-time success

• Peak Hour Failure: 12-15% (network congestion)

### D.4 Market Positioning

**Chapa Strengths**: Fastest API, developer-friendly, urban merchant adoption
**Chapa Weaknesses**: No transport workflows, no driver tools
**Telebirr Strengths**: Massive scale, nationwide coverage
**Telebirr Weaknesses**: Slow API, no QR, telecom-focused

**Proposed System Differentiation**:

1. Transport-First: Built for minibus fare collection (not retrofitted)

2. Hybrid Access: 100% coverage vs 45% smartphone limitation

3. Agent Economy: Creates revenue for many agents

4. Bureau Partnership: Exclusive dashboards for regulators

### D.5 Recommendation

**Primary Integration**: **Chapa** (faster API, transport-ready)
**Secondary**: Telebirr USSD fallback for rural routes
**Unique Value**: Agent network + driver dashboards neither competitor offers

**Sources**: Chapa Developer Portal 2025, Telebirr Annual Report Q4 2025